

SOME CALCULATIONS INVOLVING PARENTERAL ADMIXTURES

Objectives

- **Upon successful completion of parenteral admixture, you will be able to:**
- Make calculations for intravenous additives;
- Make rate-of-flow calculations for intravenous fluids and;
- Make calculations for the constitution of dry powders for parenteral use.

Injections

- ***Injections*** are sterile pharmaceutical solutions or suspensions of a drug substance in an aqueous or non- aqueous vehicle.
- They are administered by needle into almost any part of the body,
- including the joints (***intra-articular***), joint fluid (***intrasynovial***), spinal column (***intraspinal***), spinal fluid (***intrathecal***), arteries (***intra-arterial***), and in an emergency, even the heart (***intracardiac***).

Injections Cont'd

- However, most injections are administered into a vein (*intravenous, I.V., IV*), muscle (*intramuscular, I.M., IM*), skin (*intra dermal, I.D., ID, intracutaneous*), or under the skin (*subcutaneous, subQ, SQ, hypodermic*).

Injections Cont'd

- Injections are packaged in small volumes in ***ampuls*** or in prefilled disposable syringes for single-dose use; in ***vials*** and pen-injectors for single- or multiple-dose use; or in large volume plastic bags or glass containers for administration by slow intravenous *infusion*.

Injections Cont'd

- ***Parenteral*** is defined as *any medication route other than the alimentary canal* and thus includes all routes of injection.
- ***Intravenous (IV) infusions*** are sterile, aqueous preparations administered intravenously in relatively large volumes.
- They are used to extend blood volume and/or provide electrolytes, nutrients, or medications.
- Most intravenous infusions are administered to critical care, infirm, dehydrated, or malnourished patients, or to patients prior to, during, and/or following surgery.
- **Intravenous infusions are widely employed in emergency care units, in hospitals and other patient care**

Table 1: Some common intravenous infusion Solution

SOLUTION ^a	ABBREVIATION
0.9% Sodium Chloride	NS (Normal Saline)
0.45% Sodium Chloride	$\frac{1}{2}$ NS
5% Dextrose in Water	D5W or D ₅ W
10% Dextrose in Water	D10W or D ₁₀ W
5% Dextrose in 0.9% Sodium Chloride	D5NS or D ₅ NS
5% Dextrose in 0.45% Sodium Chloride	D5 $\frac{1}{2}$ NS or D ₅ 1/2NS
Lactated Ringer's (0.86% Sodium Chloride, 0.03% Potassium Chloride, 0.033% Calcium Chloride)	LR
5% Dextrose in Lactated Ringer's	D5LR or D ₅ LR

Calculations of Basic Intravenous Infusions

1. How many grams each of dextrose and sodium chloride are used to prepare a 250-mL bag of D51/2NS for intravenous infusion?

$250 \text{ mL} \times 0.05 \text{ (5\% w/v)} = 12.5 \text{ g dextrose, and}$
 $250 \text{ mL} \times 0.0045 \text{ (0.45\% w/v)} = 1.125 \text{ g sodium chloride.}$

2. Compare (a) the number of drops and (b) the length of time, in minutes, required to deliver 50-mL of intravenous solutions when using a micro drip set, at 60 drops/mL, and a standard administration set, at 15 drops/mL, if in each case one drop is to be administered per second.

Calculation cont'd

- Microdrip set:
 - (a) $60 \text{ drops/mL} * 50 \text{ mL} = 3000 \text{ drops}$
 - (b) $3000 \text{ drops} / 60 \text{ drops/minute} = 50 \text{ minutes}$
- Standard set:
 - (a) $15 \text{ drops/mL} * 50 \text{ mL} = 750 \text{ drops}$
 - (b) $750 \text{ drops} / 60 \text{ drops/minute} = 12.5 \text{ minutes}$

Calculations of Additives to Intravenous Infusion Solutions

3. A medication order for a patient weighing 154 lb. calls for 0.25 mg of amphotericin B per kilogram of body weight to be added to 500 mL of 5% dextrose injection. If the amphotericin B is to be obtained from a constituted injection that contains 50 mg/10 mL, how many milliliters should be added to the dextrose injection?

$$154 \text{ (lb.)} / 2.2 \text{ (lb.)} = 70 \text{ kg}$$

$$0.25 \text{ mg} * 70 = 17.5 \text{ mg}$$

Calculations of Additives Cont'd

- Constituted solution contains 50 mg/10 mL:
50 (mg)=10ml
17.5 (mg) ?
=3.5 mL

Rate of Flow of Intravenous Fluid

- On medication orders, the physician specifies the rate of flow of intravenous fluids in milliliters per minute, drops per minute, amount of drug (as milligrams per hour), or, more frequently, as the approximate duration of time of administration of the total volume of the infusion.
- Pharmacists may be called on to perform or check rate-of flow calculations:

Rate of Flow Cont'd

- Rate of flow (drops/minute)=Volume infusion (mL) * Drip set (drops/mL)/Time (minutes).

Rate-of-Flow Calculations Cont'd

1. A medication order calls for 1000 mL of D5W to be administered over an 8-hour period. Using an IV administration set that delivers 10 drops/mL, how many drops per minute should be delivered to the patient?

Given: Volume of fluid=1000ml

8-hours=480minutes

=1000/480=2.08ml/minute

=2.08ml/minute*10drops/ml

=20.8drop/minute

Constitution of Dry Powders

- ***Pharmaceutical compounding*** is the process by which pharmacists combine therapeutically active ingredients with pharmaceutical materials in the preparation of prescriptions and medication orders to meet the specific needs of *individual patients*.
- This is in contrast to the *large-scale* production of pharmaceutical products to meet the needs of the general population that is termed ***pharmaceutical manufacturing***.

✓ **Compounding** is an activity for which pharmacists are uniquely qualified by virtue of their education, training, and experience. Many pharmacists have developed *specialized practices* in compounding that have contributed greatly to the medication needs of their patients.

Consideration in Compounding Practices

- **Compounded prescriptions may be desired for a number of reasons, including:**
 - ✓ the need to adjust the strength or dose of a commercially available product to meet the specific requirements of a patient (e.g., a pediatric patient);
 - ✓ the need to provide a product more organoleptically acceptable (e.g., taste) to a pediatric or veterinary patient;
 - ✓ the need to prepare a different dosage form (e.g., a liquid) than the commercially available product (e.g., a tablet) to meet the requirements of a patient unable to swallow the existing dosage form (e.g., a pediatric or elderly patient);

Compounding Practices Cont'd

- ✓ the need to prepare a dosage form free of an agent (e.g., sugar, preservatives) in the commercially available product that cannot be tolerated by a patient; and
- ✓ the need to provide a patient with a specifically designed formulation of an approved drug or drug combination, which is unavailable as a commercial product.

Compounding Practices Cont'd

- *Accordingly, Compounding Process* is perform necessary calculations to establish the amounts of ingredients needed.

Constitution of Dry Powders for Oral Solution or Suspension

- Some drugs, most notably antibiotics, lose their potency in a relatively short period when prepared in a liquid dosage form.
- To enhance the shelf-life of these drugs, manufacturers provide products to the pharmacy **in dry powder form** for *constitution* with purified water or special diluent at the time a prescription or medication order is received.
- Depending on the product, the dry powder may be stable for about 24 months. After constitution, the resultant solution or suspension is stable in the quantities usually dispensed, for up to 10 days at room temperature or 14 days if maintained under refrigeration.

- Dry powders for constitution are packaged in self-contained bottles of sufficient size to accommodate the addition of the required volume of diluent (Figure 1)



(Figure 1)

Dry Powders for constitution

- On receipt of a prescription order, the pharmacist follows the label instructions for constitution, adding the proper amount of purified water or other diluent to prepare the liquid form.
- Depending on the product's formulation, constitution results in the preparation of a clear *solution* (often called a *syrup*) or a *suspension*.
- The final volume of product is the sum of the volume of solvent or diluent added and the volume occupied by the dissolved or suspended powder mixture.
- These products generally are intended for infants and children but also can be used by adults who have difficulty swallowing the counterpart solid dosage form products, such as tablets and capsules.

Dry Powder Cont'd

- For children and adults, a constituted product for oral solution or suspension generally is formulated such that the usual dose of the drug is contained in a teaspoonful amount of product.
- For infants, *pediatric drops*, which also require constitution, may be used.
- **The calculations** that follow demonstrate the preparation of constituted products and how the drug concentrations can be designed to meet a particular patient's requirements.

Calculations for the Constitution of Dry Powders for Oral Use

- 1. The label for a dry powder package of cefprozil (CEFZIL) for oral suspension directs the pharmacist to add 72 mL of purified water to prepare 100 mL of suspension. If the package contains 2.5 g of cefprozil, how many milligrams of the drug would be contained in each teaspoonful dose of the constituted suspension?*

$$\begin{aligned} 2.5 \text{ g} &= 2500 \text{ mg} \\ \frac{2500 \text{ mg}}{100 \text{ ml}} &= \frac{x \text{ mg}}{5 \text{ mL}} \\ x &= 125 \text{ mg cefprozil, answer.} \end{aligned}$$

Or, solving by dimensional analysis:

$$\frac{1000 \text{ mg}}{1 \text{ g}} \times \frac{5 \text{ mL}}{1 \text{ tsp}} \times \frac{2.5 \text{ g}}{100 \text{ mL}} = 125 \text{ mg cefprozil, answer.}$$

Calculations for the Constitution Cont'd

2. Label instructions for an ampicillin product call for the addition of 78 mL of water to make 100 mL of constituted suspension such that each 5 mL contains 125 mg of ampicillin. Calculate the volume of dry powder in the product and the total content of ampicillin.

Volume of dry powder: Because the addition of 78 mL of water results in the preparation of 100 mL of product, the volume occupied by the dry powder is:

$$100 \text{ mL} - 78 \text{ mL} = 22 \text{ mL}$$

Calculations for the Constitution Cont'd

- Total drug (ampicillin) present: If, in the constituted product, each 5 mL contains 125 mg of ampicillin, the total amount of ampicillin in the 100-mL product is:
 - 5 mL =125 mg
100 ml ?
=2500 mg

Constitution of Dry Powders for Parenteral Solution

- In contrast to the dry powders intended for oral use after constitution, injectable products may contain only limited amounts of specified added ingredients to increase the stability and effectiveness of the drug (obviously, no colorants, flavorants, or sweeteners are added).
- So, in effect, the bulk volume of the dry contents of a vial is largely or entirely the medication.

Parenteral Solution Cont'd

- If the quantity of the dry drug powder is small and does not contribute significantly to the final volume of the constituted solution, the volume of solvent used will approximate the final volume of solution.
- For example, if 1000 units of a certain antibiotic in dry form are to be dissolved, and if the powder does not account for any significant portion of the final volume, the addition of 5 mL of solvent will produce a solution containing 200 units/mL.

Parenteral Solution Cont'd

- If the dry powder, however, because of its bulk, contributes to the final volume of the constituted solution, the increase in volume produced by the drug must be considered, and this factor must then be used in calculating the amount of solvent needed to prepare a solution of a desired concentration.

Calculations for the Constitution of Dry Powders for Parenteral Use

1. Using a vial containing 200,000 units of penicillin G potassium, how many milliliters of solvent should be added to the dry powder in preparing a solution having a concentration of 25,000 units/mL?

$$\frac{25,000 \text{ (units)}}{200,000 \text{ (units)}} = \frac{1 \text{ (mL)}}{x \text{ (mL)}}$$

$x = 8 \text{ mL, answer.}$

Calculations Cont'd

2. Using a vial containing 200,000 units of penicillin G sodium and sodium chloride injection as the solvent, explain how you would obtain the penicillin G sodium needed in preparing the following prescription.

Rx:

✓ Penicillin G Sodium 15,000 units per mL

✓ Sodium Chloride Injection ad 10

✓ Sig. For IM Injection. mL

$= 15,000 \text{ units} \times 10 = 150,000 \text{ units of penicillin G sodium needed}$

Calculations Cont'd

❖ Because the dry powder represents 200,000 units of penicillin G sodium or $\frac{4}{3}$ times the number of units desired, $\frac{3}{4}$ of the powder will contain the required number of units.

Step 1. Dissolve the dry powder in 4 mL of sodium chloride injection.

Step 2. Use 3 mL of the constituted solution, *answer*.
Or, solving by dimensional analysis:

$$10 \text{ mL} \times \frac{15,000 \text{ units}}{1 \text{ mL}} \times \frac{4 \text{ mL}}{200,000 \text{ units}} = 3 \text{ mL, answer.}$$

Calculations Cont'd

3. The package information enclosed with a vial containing 5,000,000 units of penicillin G potassium (buffered) specifies that when 23 mL of a sterile solvent are added to the dry powder, the resulting concentration is 200,000 units/mL. On the basis of this information, how many milliliters of sterile water for injection should be used in preparing the following solution?

Rx Penicillin G Potassium (buffered) 5,000,000 units
Sterile Water for Injection q.s.
Make solution containing 500,000 units per mL
Sig. One mL = 500,000 units of Penicillin G Potassium

Calculations Cont'd

- ❖ The package information states that the constituted solution prepared by dissolving 5,000,000 units of the dry powder in 23 mL of sterile solvent has a final volume of 25 mL. The dry powder, then, accounts for 2 mL of this volume.

Step 1. The final volume of the prescription is determined as follows:

$$500,000(\text{units})=1(\text{mL})$$

$$5,000,000(\text{units})=?$$

$$=10\text{mL}$$

Step 2. 10 mL - 2 mL (dry powder accounts for this volume) = 8 mL, *answer*

Calculation Involving Radiopharmaceuticals

Selected Calculation Involving Radiopharmaceuticals

Objectives

Upon successful completion of this chapter, the you will be able to:

- ✓ Differentiate, quantitatively, between the standard units of radioactivity;
- ✓ Convert between the units of radioactivity;
- ✓ Make half-life calculations and;
- ✓ Perform activity and dose calculations for radiopharmaceuticals.

Radioisotopes

- The atoms of a given element are not necessarily alike. In fact, certain elements actually consist of several components, called ***isotopes***, that are chemically identical but physically may differ slightly in mass.
- Isotopes, then, may be defined as atoms that have the same nuclear charge, and hence the same atomic number, **but different masses**.
- The mass number physically characterizes a particular isotope.

Radioisotopes Cont'd

- Isotopes can be classified as **stable and unstable**.
- Stable isotopes never change unless affected by some outside force;
- unstable isotopes are distinguishable by radioactive transformations and hence are said to be **radioactive**.
- The radioactive isotopes of the elements are called ***radioisotopes*** or ***radionuclides***.
- They can be divided into two types: **naturally** occurring and **artificially** produced radionuclides.

Radioisotopes Cont'd

- The use of naturally occurring radioisotopes in medicine dates back to early in the twentieth century, when **radium** was first introduced in radiologic practice.
- It was not until after 1946, however, that artificially produced radioisotopes were readily available to hospitals and to the medical profession.
- Since that time, radionuclides have become important tools in medical research, and selected radioisotopes have been recognized as **extremely valuable diagnostic and therapeutic agents**.

Radioisotopes Cont'd

- The branch of medicine that utilizes radioisotopes and radiation in the diagnosis and treatment of disease is ***nuclear medicine***.
- Pharmacists who prepare radioactive pharmaceuticals or ***radiopharmaceuticals*** for use in patient care, practice ***nuclear pharmacy*** and are referred to as ***nuclear pharmacists***.

Radioactivity

- The breakdown of an unstable isotope is characterized by **radioactivity**.
- In the process of radioactivity, an unstable isotope undergoes changes until a stable state is reached, and in the transformation, it emits energy in the form of radiation.
- This radiation may consist of ***alpha particles***, ***beta particles***, and ***gamma rays***.
- The stable state is reached as a result of *radioactive decay*, which is characteristic of all types of radioactivity.
- Individual radioisotopes differ in the rate of radioactive decay, but in each case, a definite time is required for half the original atoms to decay.
- This time is called the ***half-life*** of the radioisotope.
- Each radioisotope, then, has a **distinct half-life**.

Radioactivity Cont'd

- The rate of decay is always a constant fraction of the total number of undecomposed atoms present.
- Mathematically, the rate of disintegration may be expressed as follows:

$$dN/dt = \lambda N$$

in which N is the number of undecomposed atoms at time t , and λ is the decay constant or the fraction disintegrating per unit of time.

- The constant may be expressed in any unit of time, such as reciprocal seconds, minutes, or hours, among others.
- The numeric value of the decay constant will be 24 times as great when expressed in days, for example, as when expressed in hours.

Radioactivity Cont'd

- Integrated: $N = N_0 e^{-\lambda t}$

in which N is the number of atoms remaining at elapsed time t , N_0 is the number of atoms originally present (when $t=0$), λ is the decay constant for the unit of time in terms of which the interval t is expressed, and e is the base of the natural logarithm 2.71828.

HALF-LIVES OF SOME RADIOISOTOPES

RADIOISOTOPE	HALF-LIFE
^{198}Au	2.70 days
^{14}C	5,700 years
^{45}Ca	180 days
^{57}Co	270 days
^{60}Co	5.27 years
^{51}Cr	27.8 days
^{59}Fe	45.1 days
^{67}Ga	78.3 hours
^{203}Hg	46.6 days
^{123}I	13.2 hours
^{125}I	60 days
^{131}I	8.08 days
^{111}In	2.83 days
^{42}K	12.4 hours
$^{81\text{m}}\text{Kr}$	13.1 seconds
^{99}Mo	2.6 years
^{22}Na	2.6 years
^{24}Na	15.0 hours
^{32}P	14.3 days
^{35}S	87.2 days
^{75}Se	120 days
^{85}Sr	64 days
$^{99\text{m}}\text{Tc}$	6.03 hours
^{201}Tl	73.1 hours
^{133}Xe	5.24 days
^{169}Yb	32.0 days

Radioactivity Cont'd

- Because the rate of decay can also be characterized by the half-life ($T_{1/2}$), the value of N in the integration equation at the end of a half period is $1/2N_0$
- The equation then becomes,

$$1/2N_0 = N_0^{-\lambda t_{1/2}}$$

Solving by Natural logarithm

$$\ln 1/2 = -\lambda T_{1/2}$$

$$\text{Then } \lambda T_{1/2} = 2.303 \log 2$$

$$T_{1/2} = 0.693/\lambda$$

Units of Radioactivity

- The quantity of activity of a radioisotope is expressed in absolute units (total number of atoms disintegrating per unit time).
- The basic unit is the **curie** (Ci), which is defined as that quantity of a radioisotope in which 37 billion atoms disintegrate per second.
- The **millicurie** (mCi) is one thousandth of a curie, and the **microcurie** (μCi) is one millionth of a curie. The **nanocurie** (nCi), also known as the **millimicrocurie**, is one billionth of a curie (10^{-9} Ci).

Calculations of Half-Life and Disintegration Constant

1. *The disintegration constant of a radioisotope is 0.02496 day⁻¹. Calculate the half-life of the radioisotope.*

✓ $T_{1/2} = 27.76$ or 27.8 days, *answer.*
2. *The half-life of ^{198}Au is 2.70 days. Calculate its disintegration constant.*
 $\lambda = 0.2567 \text{ day}^{-1}$

Quiz *

1. How many grams each of sodium chloride and dextrose are present in a 1000- mL IV bag of 0.18% sodium chloride and 4% dextrose?
2. A patient received a 500-mL whole blood transfusion at 9:13 PM at a rate of 30 drops per minute via an administration set that delivers 10 drops/mL. At what time was the transfusion completed?
3. A patient received 500 mL of D5W1/2NS at a rate of 15 drops/minute. If the administration set used delivered 15 drops/mL, calculate the infusion time in hours and minutes.

Quiz ** Cont'd

4. A physician orders 2 g of an antibiotic to be placed in 1000 mL of D5W. Using an injection that contains 300 mg of the antibiotic per 2 mL, how many milliliters should be added to the dextrose injection in preparing the medication order?
5. *The original quantity of a radioisotope is given as $500\mu\text{Ci/mL}$. If the quantity remaining after 16 days is $125\mu\text{Ci /mL}$, calculate (a) the disintegration constant and (b) the half life of the radioisotope.*

Thank you !